ANALYSIS OF ORDER DYNAMICS IN THE ELECTRICITY SPOT MARKET: A MULTIVARIATE & FUNCTIONAL PERSPECTIVE

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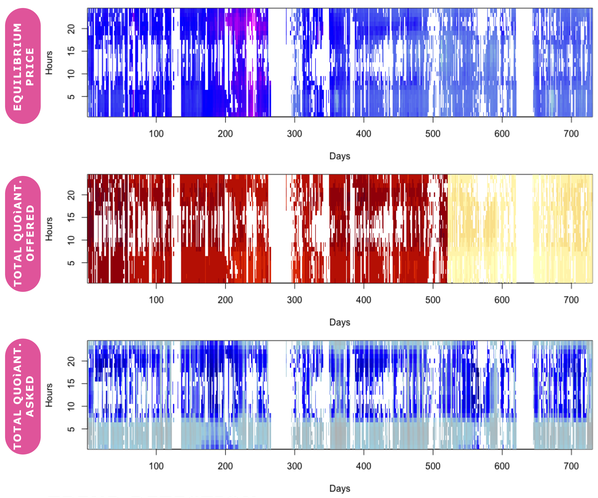
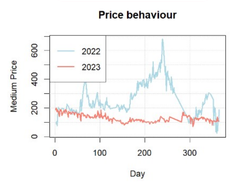
THE GOAL

Exploiting the demand and supply curves through multivariate and functional statistical techniques the purpose of the project is to **investigate the dynamics** which characterize the electricity market, and to make prediction of the curves within this highly volatile enviroment.

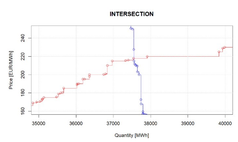
DATA PREPROCESSING & EXPLORATORY ANALYSIS



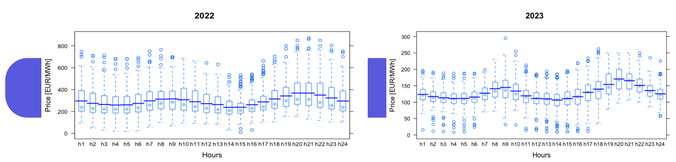
FIXING AN HOUR



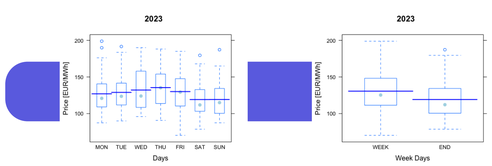
FIXING A DAY AND AN HOUR



TREND DETECTION:



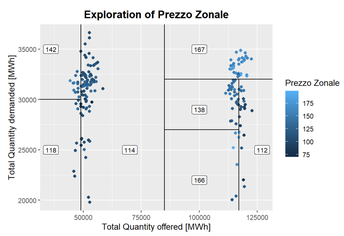
GROUPING BY HOUR



GROUPING BY DAYS OF THE WEEK

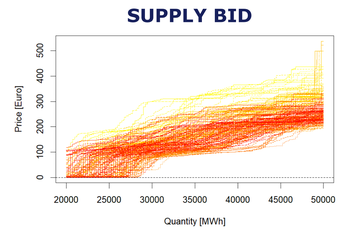
CART INTERPRETATION:

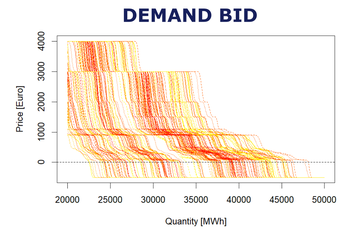
- For high quantity of energy producted the final price is low, regardless of the demand quantity
- When the total quantity of energy producted is low, the equilibriums price seems to be higher, especcially when the demand is higher

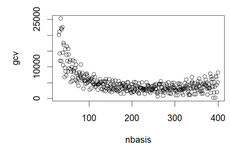


FUNCTIONAL DATA ANALYSIS

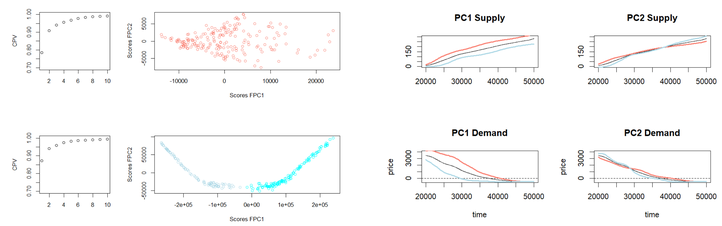
B-SPline smoothing with:
- Range: 20000 MWh - 50000 MWh
- Zero order spline (m=1)
- Number of basis: 120
- First FPCA in order to treat the outliers





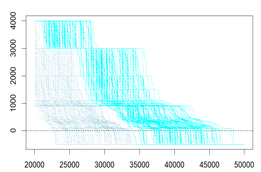


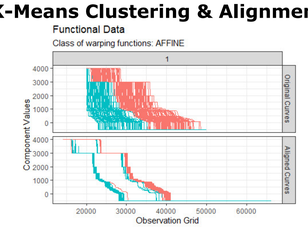
Example of a GCV plot



The **FPCA** shows that:
- The interpretation of the first 2 PC's is the same
- The supply curves are well spread in the 2D-space
- The demand cures can be clustered in two groups

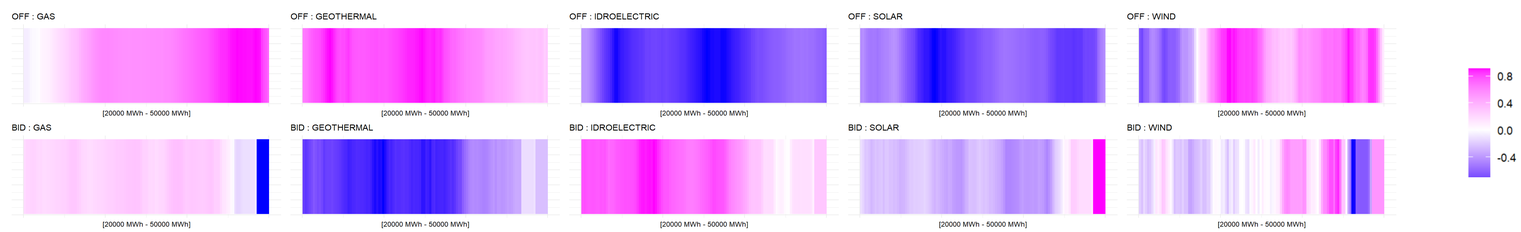
Hierarchical Clustering vs **K-Means Clustering & Alignment**





Hclust and K-Means provide the same distinction: weekend and working days
The warping shows that the majority of the curves within a cluster are similar.

CORRELATION WITH EXTERNAL VARIABLES



PREDICTION ATTEMPT

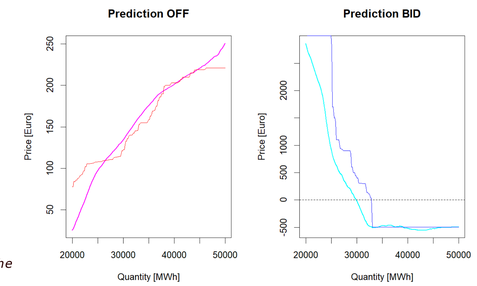
FAR(1): Functional Auto-Regressive Model of order 1

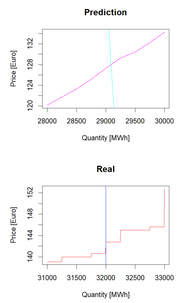
$$X_{n+1} = \Psi(X_n) + \varepsilon_{n+1}, \quad \text{where } \Psi \text{ is an Hilbert-Schmidt operator}$$

$$\Psi(X)(t) = \int \psi(t,s)X(s)ds, \quad \text{and } \psi \text{ is a bivariate kernel}$$

We estimate ψ through the **Estimated Kernel Method**

Reference: D. Didericksen, P. Kokoszka, X. Zhang in *"Empirical properties of forecasts with the functional autoregressive model"* (2011)



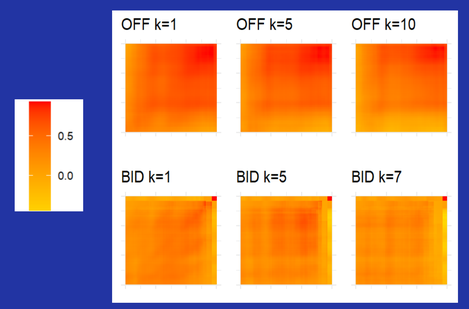


Pred Equilibrium Price:
129 [Euro]

Real Equilibrium Price:
141 [Euro]

CONCLUSIONS & POSSIBLE IMPROVMENTS

- There are clear trends and usual pattern during the days
- The supply bid curves are highly correalted among them and tends to be similar
- The demand bid curves are way less correlated and more can be clearly clustered into groups, characterized by the day of the week
- The external variables affect in a significant way the supply bid curve, while in a less effective way the demand curve



The prediction part can be improved building a more complex model. Other intristing approaches can be:
- FAR(1) with Predictive Factor Method
- FAR(k) and a Black-Box Model Identification
- Taking into account the clustering structure and the correlation with external variables

Datasets:
Electricity & Gas: www.MercatoElettrico.org
External Variables: www.transparaency.entsoe.eu